

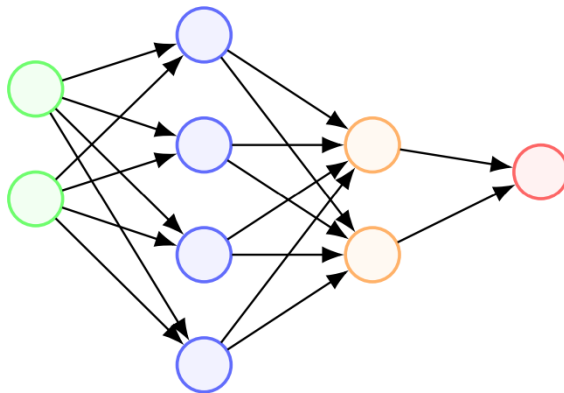
Quiz 1

Student Name: _____

Points: 30

A Number: _____

1. Consider a Neural Network diagram:



If we are supplying two samples with two features (green bubbles), then what is the dimension of W s responsible for compressing 4 meta features to 2 meta features? For this assume that features in a feature matrix are along the column dimensions, and samples are along row dimensions.

- (a) $W_{4 \times 2}$ ✓
 - (b) $W_{2 \times 4}$
 - (c) $W_{4 \times 4}$
 - (d) $W_{2 \times 2}$
2. Consider an activation function as $\sigma(\mathbf{x}) = \frac{1}{1+e^{-\mathbf{x}}}$, if input is $\mathbf{x} = [1.09, 2.01, 3.0]$, the what is the dimension of $A = \sigma(\mathbf{x})$?
- (a) 3×3
 - (b) 1
 - (c) Cannot be determined
 - (d) 3 ✓
3. What does `curl` command do in a Linux machine?
- (a) It installs Python.
 - (b) It searches websites for data.

```

0 0 0 0 1 1 1 1 0 0 0 0 1 1 0 1 1 1 0 0 0 1
1 0 0 1 0 1 0 0 0 1 0 0 1 1 0 0 1 0 1 1
0 0 1 1 0 0 1 0 1 1 1 0 1 1 0 1 0 1 0 0 0

```

- (c) It transfers data from server to client. ✓
- (d) It creates a python virtual environment.
4. Which Linux command is used for create a new directory?
- (a) cd
- (b) mkdir ✓
- (c) ls
- (d) mv

5. The function

$$g(z) = \begin{cases} 1, & z \geq \theta \\ 0, & \text{otherwise} \end{cases}$$

is fully differentiable.

- (a) True
- (b) False ✓
6. Maximum value of the Sigmoid function $\sigma(\mathbf{x}) = \frac{1}{1+e^{-x}}$ is
- (a) -1
- (b) +1 ✓
- (c) 0
- (d) 0.5

7. Consider a Neural Network definition:

```

1 class QuizModel(nn.Module):
2     def __init__(self):
3         super().__init__()
4         self.fc1 = nn.Linear(10, 5)
5         self.relu = nn.ReLU()
6         self.fc2 = nn.Linear(5, 2)
7
8     def forward(self, x):
9         x = self.fc1(x)
10        x = self.relu(x)
11        x = self.fc2(x)
12        return x

```

What is the size of the final output?

- (a) 2 ✓

```

0 0 0 0 1 1 1 1 0 0 0 0 1 1 0 1 1 1 0 0 0 1
1 0 0 1 0 1 0 0 0 0 2 0 0 1 1 0 0 1 0 1 1
0 0 1 1 0 0 1 0 1 1 1 0 1 1 0 1 0 1 0 0 0

```

- (b) 5
- (c) 1
- (d) 2×5

8. Consider

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 6 \end{bmatrix}$$

and

$$\mathbf{y} = \mathbf{f}(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ x_1 + x_2 \\ x_1 x_2 + x_2 x_3 \end{bmatrix}$$

What is the size of Jacobian J ?

- (a) 4×4
- (b) 4×3
- (c) 3×4 ✓
- (d) 3×3

9. Dropout

- (a) randomly ignores a few neurons to create subnetwork on which training happens ✓
- (b) randomly ignores a few samples to train on a subset of data samples
- (c) randomly ignores a few weights to train using remaining weights
- (d) randomly ignores a few layers to train using remaining layers

10. A higher error on test set compared to training set denotes

- (a) A lack of generalizability of the trained model
- (b) Underfitting
- (c) Overfitting
- (d) Both (A) and (C) ✓